

McMaster Micromouse Competition Rules

2026

General Overview

Welcome to the 2026 McMaster Micromouse Competition! The objective is to build an autonomous robot (a “mouse”) that can independently navigate a maze to find the destination in the shortest time possible.

- Teams may consist of one to five members.
- These rules are adapted from the official IEEE Micromouse specifications with minor local adjustments.
- **Note: Terms highlighted in red are provisional. Final values and parameters may be adjusted based on event logistics and registration numbers.**

Robot Specifications

- The robot must be designed and built from scratch by the competing team members.
- The robot must be entirely self-contained and autonomous. Remote controls, tethers, or external communication devices are prohibited.
- The robot’s physical footprint must remain within a 25 cm x 25 cm area at all times. There are no height restrictions.
- The robot cannot separate or leave any parts of itself behind while navigating. The entire robot must arrive at the destination square as one unit to successfully complete the maze.
- The robot must not jump over, fly above, climb, scratch, cut, burn, mark, damage, or otherwise destroy the walls or floor of the maze.

Maze Specifications

- The maze is laid out on a **16 x 16** unit grid. Each individual unit square measures 18 cm x 18 cm. The entire maze is securely enclosed by a perimeter wall.
- The walls feature white sides and red tops. The maze floor is painted black to aid in sensor contrast.
- The starting position is located at one of the four corners of the maze.

Competition & Run Rules

- Each team is allotted **10 minutes** inside the maze. During this time, the robot may make multiple autonomous runs to explore, map the maze, and execute fast speed-runs.
- The official run time is tracked using a start line (exiting the starting square into the next unit square) and a finishing line (the entrance threshold to the destination square). The team's best (shortest) run time will be recorded as their final score.
- Information regarding the specific maze layout CANNOT be downloaded, programmed into, or given to the robot beforehand. It must map the maze dynamically.
- Once the maze is revealed, teams may physically touch their robot between runs to make the following adjustments:
 1. Changing switch/button settings (e.g., selecting different search algorithms)
 2. Replacing or recharging batteries
 3. Adjusting physical sensor alignments
 4. Changing speed profile settings
 5. Performing minor mechanical repairs
- Teams may safely abort and restart their runs at any time. If the robot becomes immobilized, stuck in a corner, or caught on a wall, the team may manually intervene to rescue the robot. However, they may not alter the robot's intended direction of movement to assist its navigation.

Scoring & Awards

- **Top Placements:** First, second, and third place will be awarded to the teams achieving the shortest official run times. *Note: To qualify for these top placements, the robot must complete its winning run continuously, without the team touching or intervening with the robot during that specific run.*
- **Honourable Mentions:** We will also recognize teams for creativity and engineering with honourable mentions in the following categories:
 - **[TBD]**

Good luck and have fun everyone!